ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

MACHINE VISION

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 3rd year final examination: image formation and cameras, digital images and point operations, filtering and edge detection, binary images, morphology and features, stereo vision and vision systems for industrial inspection and robot guidance.

Written by Academic Central for practice, following undergraduate machine vision and image processing courses for mechatronics students (topics as in the image processing and robotic vision content of South African engineering programmes, e.g. University of Pretoria EAI 733, and standard introductory machine vision courses). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Show ALL your working. Pixel positions are given as (row, column), counting from 0 at the top left.
3. Non-programmable scientific calculators may be used. Give answers correct to TWO decimal places.
4. Number your answers exactly as the questions are numbered.

QUESTION 1: IMAGE FORMATION AND CAMERAS

1.1 A camera has a lens with a focal length of 8 mm. An object 0.5 m tall stands 2 m in front of it.

1.1.1 Use the pinhole model to calculate the height of its image on the sensor. (2)

1.1.2 The pixels are 4 µm square. How many pixels tall is the image? (2)

1.2 A camera has f_x = f_y = 800 pixels and its principal point at (c_x; c_y) = (320; 240). Project the point (X; Y; Z) = (0.2; −0.1; 2) m, given in the camera frame, to pixel coordinates (u; v). (4)

1.3 A sensor is 6.4 mm wide and the lens has f = 8 mm. Calculate the horizontal field of view. (3)

1.4 A camera must see defects as small as 1 mm across a field 400 mm wide, with at least 2 pixels across the smallest defect. Calculate the number of pixels needed across the image. (3)

1.5 What is radial lens distortion, and why is a camera calibrated before measurements are made? (3)

1.6 Explain why a global-shutter camera is chosen for parts moving fast on a conveyor, rather than a rolling-shutter camera. (3)

[20]

QUESTION 2: DIGITAL IMAGES AND POINT OPERATIONS

2.1 Calculate the memory needed for one 640 × 480 colour image with 8 bits per channel (RGB). (2)

2.2 Convert the RGB pixel (200; 100; 50) to grey using Y = 0.299R + 0.587G + 0.114B. (2)

2.3 The grey levels in a dark image range from 50 to 150. Write the linear contrast-stretching function that maps this range to 0 to 255, and apply it to a pixel of value 100. (3)

2.4 Explain how Otsu's method chooses a threshold automatically. (3)

2.5 Apply a threshold of 128 to the image below (pixel ≥ 128 becomes 1, otherwise 0), and write down the binary image. (3)

12	140	200	90
130	220	210	60
15	180	127	20

10	30	128	255

2.6 Why is the HSV colour space often better than RGB for finding a coloured object under changing lighting? (2)

[15]

QUESTION 3: FILTERING AND EDGE DETECTION

Answer 3.1 to 3.3 using the 5 × 5 grey image I below. It has a vertical edge between columns 2 and 3, and one noisy pixel at (2, 0).

10	10	10	80	80
10	10	10	80	80
250	10	10	80	80
10	10	10	80	80
10	10	10	80	80

3.1 Calculate the output of a 3 × 3 mean filter at pixel (2, 2). (3)

3.2 At pixel (2, 1):

3.2.1 Calculate the output of a 3 × 3 median filter. (3)

3.2.2 Calculate the output of a 3 × 3 mean filter, and say which filter is better for this kind of noise. (2)

3.3 The Sobel kernels are G_x = [−1 0 1 ; −2 0 2 ; −1 0 1] and G_y = [−1 −2 −1 ; 0 0 0 ; 1 2 1]. At pixel (1, 2):

3.3.1 Calculate G_x. (4)

3.3.2 Calculate G_y. (2)

3.3.3 Calculate the gradient magnitude and direction, and explain what they show. (2)

3.4 Why is an image usually smoothed with a Gaussian filter before edges are detected? (2)

3.5 List the main steps of the Canny edge detector. (4)

3.6 Briefly explain how the Hough transform finds straight lines in an edge image. (3)

[25]

QUESTION 4: BINARY IMAGES, MORPHOLOGY AND FEATURES

4.1 Define erosion, dilation, opening and closing, and give a use for opening and for closing. (6)

4.2 The 5 × 5 binary image below has a 3 × 3 square object and one noise pixel.

1	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

4.2.1 Write down the result of eroding it with a 3 × 3 square structuring element (pixels outside the image count as 0). (3)

4.2.2 Write down the result of an opening (erosion followed by dilation) with the same element, and say what it achieved. (2)

4.3 Explain connected-component labelling, and the difference between 4-connectivity and 8-connectivity. (3)

4.4 A blob consists of the pixels (1, 1), (1, 2), (2, 1), (2, 2) and (3, 2). Calculate its area and its centroid. (4)

4.5 Calculate the circularity $4\pi A / P^2$ of a part with an area of 1 200 pixels and a perimeter of 150 pixels. What value would a perfect circle give? (2)

[20]

QUESTION 5: VISION SYSTEMS AND 3D

5.1 Design a vision system that checks whether every bottle on a conveyor has a cap. Describe the lighting, camera, triggering and the image-processing steps, and how the result is sent to a reject mechanism. (6)

5.2 Explain template matching with normalised cross-correlation, and give TWO limitations. (3)

5.3 A stereo camera pair has a baseline of 0.12 m and a focal length of 700 pixels.

5.3.1 Calculate the depth of a point with a disparity of 35 pixels. (3)

5.3.2 Calculate the depth if the disparity is wrong by ±1 pixel, and comment. (2)

5.4 A camera above a conveyor is calibrated at 0.5 mm per pixel, and the robot's pick point is at pixel (row 240, column 320). A part's centroid is at (row 300, column 400). Calculate the offset in mm that the robot must move along the column (x) and row (y) directions. (3)

5.5 Give ONE advantage and ONE disadvantage of a deep-learning (CNN) approach compared with classical image processing for inspection. (3)

[20]

TOTAL: 100

END OF QUESTION PAPER